process, the net change in charge will always be zero.

(iii) Quantisation of charges

What is the smallest amount of charge that can be found in nature? Experiments show that the charge on an electron is $-e$ and the charge on the proton is $+e$. Here, e denotes the fundamental unit of charge. The charge q on any object is equal to an integral multiple of this fundamental unit of charge e.

$$q = ne \qquad (1.1)$$

Here n is any integer (0, ±1, ±2, ±3, ±4……..). This is called quantisation of electric charge.

Robert Millikan in his famous experiment found that the value of $e = 1.6 \times 10^{-19}$ C. The charge of an electron is -1.6×10^{-19} C and the charge of the proton is $+1.6 \times 10^{-19}$ C. When a glass rod is rubbed with silk cloth, the number of charges transferred is usually very large, typically of the order of 10^{10}. So the charge quantisation is not appreciable at the macroscopic level. Hence the charges are treated to be continuous (not discrete). But at the microscopic level, quantisation of charge plays a vital role.

EXAMPLE 1.1

Calculate the number of electrons in one coulomb of negative charge.

Solution

According to the quantisation of charge,

$$q = ne$$

Here $q = 1$C. So the number of electrons in 1 coulomb of charge is

$$n = \frac{q}{e} = \frac{1C}{1.6\times10^{-19}} = 6.25\times10^{18} \text{ electrons}$$

1.2 COULOMB'S LAW

In the year 1786, Coulomb deduced the expression for the force between two stationary point charges in vacuum or free space. Consider two point charges q_1 and q_2 at rest in vacuum, and separated by a distance of r as shown in Figure 1.2. According to Coulomb, the force on the point charge q_2 exerted by another point charge q_1 is

$$\vec{F}_{21} = k\frac{q_1 q_2}{r^2}\hat{r}_{12} \qquad (1.2)$$

where $\hat{r}_{12}$ is the unit vector directed from charge q_1 to charge q_2 and k is the proportionality constant.

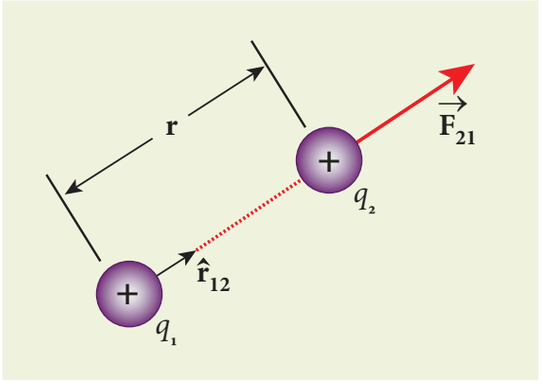


Figure 1.2 Coulomb force between two positive point charges

Important aspects of Coulomb's law

(i) Coulomb's law states that the electrostatic force is directly proportional to the product of the magnitude of the two point charges and is inversely proportional to the square of the distance between the two point charges.

(ii) The force on the charge q_2 exerted by the charge q_1 always lies along the line

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